

Microprocessors and Microcontrollers

Assignment 11- Week 11

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15 M

QUESTION 1:

Find the contents of control word of the 8255 for the following configuration:

All the ports of A, B and C are input ports (mode 1)

- a) 3Fh
- b) BFh
- c) 9Fh
- d) 1Fh

Correct Answer: b

Detailed Solution:

Understanding the 8255 Control Word

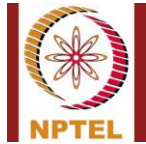
The Intel 8255 Programmable Peripheral Interface (PPI) is used for interfacing I/O devices. It has three 8-bit ports:

- Port A (PA)
- Port B (PB)
- Port C (PC) (which is further divided into upper 4 bits PC4-PC7 and lower 4 bits PC0-PC3)

The control word determines whether these ports act as input or output and in which mode they operate.

Control Word Format (Mode Set Register)

The control word is an 8-bit binary value, formatted as follows:



Bit	Function
D7	Control Word Set (Always 1)
D6	Port A: 1 = Input, 0 = Output
D5	Mode Selection for Port A
D4	Port C Upper: 1 = Input, 0 = Output
D3	Mode Selection for Port B
D2	Port B: 1 = Input, 0 = Output
D1	Port C Lower: 1 = Input, 0 = Output
D0	Mode Set Flag (Always 1)

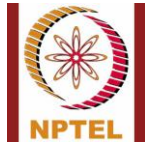
Step 1: Given Configuration

- Port A = Input (Mode 1)
- Port B = Input (Mode 1)
- Port C = Input (Both Upper & Lower)
- Mode 1 is used for Port A and Port B

Since we are using Mode 1, we need to set the appropriate bits accordingly.

Step 2: Assigning Bits Based on the Configuration

Now, let's fill in the control word bit by bit.



Bit	Function	Value	Explanation
D7	Mode Set Flag	1	Always 1 for Control Word
D6	Port A as Input	1	PA is an input port
D5	Mode selection for PA (Mode 1)	1	Mode 1 is selected for PA
D4	Port C Upper (PC4-PC7) as Input	1	PC4-PC7 as input
D3	Mode selection for PB (Mode 1)	1	Mode 1 is selected for PB
D2	Port B as Input	1	PB is an input port
D1	Port C Lower (PC0-PC3) as Input	1	PC0-PC3 as input
D0	Mode Set Operation	1	Always 1

Thus, the control word is:

1 0 1 1 1 1 1 1 = **BFh**

QUESTION 2:

Correct Answer: c

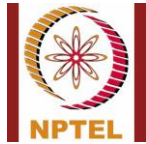
Detailed Solution:

Bit addressability refers to the ability to access and manipulate individual bits within a port rather than working with the entire 8-bit port at once.

In the Intel 8255 Programmable Peripheral Interface (PPI), only Port C has bit-addressability. This means that individual bits of Port C can be set or cleared independently without affecting the other bits of the port.

Why is Port C Bit-Addressable?

- Port C is divided into two 4-bit groups: PC0-PC3 (Lower) and PC4-PC7 (Upper).
- These bits can be individually controlled using the Bit Set/Reset (BSR) mode of the 8255.
- Ports A and B do not support bit-addressability; they can only be accessed as full 8-bit ports.



QUESTION 3:

Correct Answer: b

Detailed Solution:

AVR microcontrollers, such as the ATmega series, feature multiple timers/counters of different sizes. The most common timers are:

1. Timer/Counter 0 → 8-bit
2. Timer/Counter 1 → 16-bit
3. Timer/Counter 2 → 8-bit

These timers operate as up counters by default but can also function in different modes (e.g., PWM, CTC, etc.).

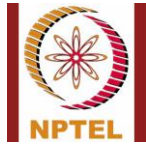
Analyzing the Given Options:

- Option (a): Incorrect → Timer 1 is 16-bit, not 8-bit.
- Option (b): Correct → Timer 0 is 8-bit, and Timer 1 is 16-bit, both acting as up counters.
- Option (c): Incorrect → Timer 1 does not function as a down counter by default.
- Option (d): Incorrect → Timer 0 is 8-bit, not 16-bit.

QUESTION 4:

Which of the following statements correctly describes the output compare feature of AVR microcontrollers?

- a) An AVR Timer/Counter 0 has a single output compare register (OCR0A).
- b) An AVR Timer/Counter 0 has dual output compare registers (OCR0A and OCR0B).
- c) An AVR Timer/Counter 1 has a single output compare register (OCR1A).
- d) An AVR Timer/Counter 1 does not support output compare registers.



Correct Answer: b

Detailed Solution:

QUESTION 5:

Correct Answer: c

Detailed Solution:

In AVR microcontrollers, the Clock Select (CS) bits (CS02, CS01, CS00) in the TCCR (Timer/Counter Control Register) determine the prescaler value for the timer/counter.

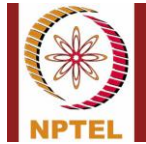
The prescaler determines how much the CPU clock is divided before feeding it to the timer. The prescaler values based on the CS bits are as follows:

CS02	CS01	CS00	Prescaler (Division Factor)
0	0	0	Timer Stopped
0	0	1	CK/1 (No Prescaler)
0	1	0	CK/8
0	1	1	CK/64
1	0	0	CK/256
1	0	1	CK/1024

Given Conditions:

- CS02 = 0
- CS01 = 1
- CS00 = 1

From the table, the corresponding prescaler value is CK/64, meaning the CPU clock is divided by 64 before being applied to the timer.



Evaluating the Answer Choices:

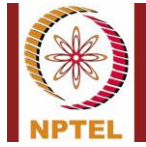
- (a) Stop, the timer/counter is stopped → Incorrect (The timer is stopped only when $CS02 = 0$, $CS01 = 0$, $CS00 = 0$).
- (b) CPU frequency is divided by 8 ($CK/8$) → Incorrect ($CK/8$ occurs when $CS02 = 0$, $CS01 = 1$, $CS00 = 0$).
- (c) CPU frequency is divided by 64 ($CK/64$) → Correct (Matches our condition: $CS02 = 0$, $CS01 = 1$, $CS00 = 1$).
- (d) CPU frequency is divided by 256 ($CK/256$) → Incorrect ($CK/256$ occurs when $CS02 = 1$, $CS01 = 0$, $CS00 = 0$).

QUESTION 6:

Correct Answer: b

Detailed Solution:

1. TCCR0 (Timer/Counter Control Register 0)
 - This is the control register for Timer/Counter 0.
 - It configures the timer mode, prescaler, and other settings.
 - Incorrect (It does not hold the counter value).
 2. TCNT0 (Timer/Counter Register 0)
 - This is the counter register for Timer/Counter 0.
 - It holds the current value of the 8-bit Timer/Counter 0.
 - Correct Answer!
 3. TCCR1A (Timer/Counter 1 Control Register A)
 - This is a control register for Timer/Counter 1, not Timer 0.
 - Incorrect (It does not hold the counter value).
 4. TCNT1 (Timer/Counter Register 1)
 - This holds the counter value for Timer/Counter 1, which is 16-bit.
 - Incorrect (It is for Timer 1, not Timer 0).
-



QUESTION 7:

Correct Answer: d

Detailed Solution:

Interrupts in an AVR microcontroller allow the CPU to respond immediately to certain events without continuously polling for them. They are used for various critical tasks, including:

1. RESET
 - A reset interrupt is triggered when the microcontroller is powered on or when an external reset is applied.
 - It ensures the system starts executing from a known state.
2. Timers and Time-Critical Code
 - Timer interrupts are widely used for tasks such as delays, PWM generation, and event scheduling.
 - They allow execution of precise, time-sensitive code without blocking the main program.
3. Hardware Signaling
 - Interrupts can be triggered by external hardware events, such as button presses, sensor inputs, and serial communication signals.
 - This allows efficient data handling without constantly checking input pins.

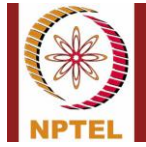
QUESTION 8:

Why are interface chips essential in a microcontroller-based system?

- a) They help in synchronizing data transfers and resolving speed mismatches between the CPU and I/O devices.
- b) They only help in resolving the speed mismatch problem.
- c) They only help in synchronizing data transfers.
- d) None of the above.

Correct Answer: a

Detailed Solution:



Interface chips are crucial in microcontroller-based systems because they serve two main purposes:

1. Synchronizing Data Transfers:
 - I/O devices and the CPU often operate at different speeds and protocols.
 - Interface chips act as a bridge to ensure smooth communication.
 2. Resolving Speed Mismatch:
 - The CPU is much faster than many peripheral devices.
 - Interface chips buffer and regulate data flow to match the speed of both ends.
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QUESTION 9:

Correct Answer: c

Detailed Solution:

Contact bouncing occurs when a switch is pressed or toggled, causing unintended multiple rapid contacts before settling into a steady state.

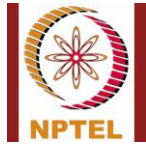
- Push-button switches: Highly prone to bouncing because of their mechanical nature when pressed or released.
 - Toggle switches: While they can also experience bouncing, it is generally less frequent than in push-buttons.
-

QUESTION 10:

Correct Answer: c

Detailed Solution:

Switch bounce occurs when mechanical contacts in a switch make multiple unintended connections before settling. This can cause false triggering in digital circuits.



One effective hardware solution is an RC (Resistor-Capacitor) debounce circuit, which smooths out the fluctuations by filtering out rapid voltage changes.

- The RC time constant should be larger than the duration of the switch bounce to allow sufficient filtering.
- LC circuits (Inductor-Capacitor) are not typically used for debouncing due to their oscillatory nature.

QUESTION 11:

Correct Answer: c

Detailed Solution:

DIP stands for Dual In-line Package.

A DIP switch is a set of small switches in a dual in-line package, often used for configuration settings in electronic circuits. These switches are manually toggled to set options without the need for software programming.

- "Dual" refers to the two parallel rows of pins.
- "In-line" means the pins are arranged in a straight line along both sides.
- "Package" refers to the physical form of the component.

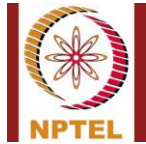
QUESTION 12:

In the 8255 Programmable Peripheral Interface (PPI), which control word should be used to reset BIT 4 of Port C in Bit Set/Reset (BSR) Mode?

- a) 00001111
- b) 10001111
- c) 10001000
- d) 00001000

Correct Answer: d

Detailed Solution:



Step 1: Understanding the 8255 BSR Mode

- The 8255 PPI has two operational modes:
 1. I/O Mode – Controls full ports (A, B, and C).
 2. Bit Set/Reset (BSR) Mode – Controls individual bits of Port C.
- In BSR mode, the control word format is:

D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0

0 | 0 | 0 | 0 | B3 | B2 | B1 | B0

- - D7 = 0 → Indicates BSR mode.
 - D6 - D4 = 000 → Reserved for BSR mode.
 - D3 - D1 → Selects the bit in Port C to be modified.
 - D0 → 0 = Reset, 1 = Set.

Step 2: Setting the Control Word

- We need to reset bit 4 of Port C.
- In binary, Bit 4 is represented as 100 (D3-D1).
- To reset it (D0 = 0), the control word becomes:

0000 1000 (Binary) = 08H (Hex)

Option (a) 00001111 → Incorrect

D7 = 0 → Correct (indicating BSR mode).

D3-D1 = 111 → Incorrect (Selects Bit 7 instead of Bit 4).

D0 = 1 → Incorrect (Sets the bit instead of resetting it).

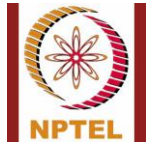
Why incorrect?

- This control word sets Bit 7 of Port C instead of resetting Bit 4.

Option (b) 10001111 → Incorrect

D7 = 1 → Incorrect (This is NOT a BSR mode instruction; it's an I/O Mode instruction).

D3-D1 = 111 → Correct if BSR mode was used, but it's not.



D0 = 1 → Correct for setting, but again, BSR mode is not active.
Why incorrect?

- D7 should be 0 to indicate BSR mode, but here it is 1, meaning this control word is for I/O mode, not BSR mode.

Option (c) 10001000 → Incorrect

D7 = 1 → Incorrect (Not a BSR mode instruction, it's an I/O mode instruction).

D3-D1 = 100 → Correctly selects Bit 4 of Port C.

D0 = 0 → Correctly resets the bit.

Why incorrect?

- D7 is 1, which means this control word is in I/O mode instead of BSR mode.
- For BSR mode, D7 must be 0.

QUESTION 13:

A common anode 7-segment display is connected to an 8051 microcontroller. If the binary combination 11000000 is applied to Port 1 (P1.7 - P1.0), which digit will be displayed?

- a) 0
- b) 1
- c) 8
- d) 6

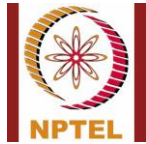
Correct Answer: a

Detailed Solution:

In a common anode 7-segment display, all anodes are connected to Vcc (logic 1), and the segments are activated by applying logic 0 (LOW) to turn them ON.

Step 1: Understanding the 7-segment LED Segments

A 7-segment display has segments labeled a, b, c, d, e, f, g.



Segment	a	b	c	d	e	f	g
0 ON/OFF	0	0	0	0	0	0	1

The binary combination given is 11000000.

- Here, 1 means the segment is OFF.
- 0 means the segment is ON.

Step 2: Decoding the Binary Pattern

11000000 corresponds to:

- a = 0 (ON)
- b = 0 (ON)
- c = 0 (ON)
- d = 0 (ON)
- e = 0 (ON)
- f = 0 (ON)
- g = 1 (OFF)

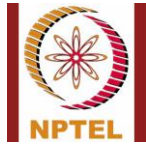
Step 3: Identifying the Digit

From the standard 7-segment display encoding, when a, b, c, d, e, f are ON and g is OFF, it represents the digit "0".

QUESTION 14:

Correct Answer: c

Detailed Solution: The 8255 PPI has three ports (Port A, Port B, and Port C) and a control register. The selection of these ports is determined by the address lines A1 and A0, while the Chip Select (CS) signal enables or disables the chip.



8255 Port Selection Table

CS	A1	A0	Selected Register
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register

Key Observations:

- CS = 0 → Chip is enabled.
- A1 = 1, A0 = 0 → Selects Port C.

QUESTION 15:

In ADC 0804, which of the following statements is incorrect?

- a) The Chip Select (CS) pin is active high.
- b) The data output pins (D7-D0) provide an 8-bit digital output.
- c) The default operating voltage range of ADC 0804 is 0 - 5V.
- d) The reference voltage can be adjusted to modify the input range.

Correct Answer: a

Detailed Solution:

Option (a): Incorrect → CS (Chip Select) is active LOW, not HIGH.

This means ADC 0804 is enabled when CS = 0, not CS = 1.

Option (b): Correct → D7-D0 are output pins that provide an 8-bit digital equivalent of the analog input voltage.

Option (c): Correct → By default, ADC 0804 operates within 0 - 5V unless a reference voltage is applied.

Option (d): Correct → The reference voltage ($V_{ref}/2$) can be adjusted to change the input voltage range.



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